Area between 2 curves

Exam code: 9MA0

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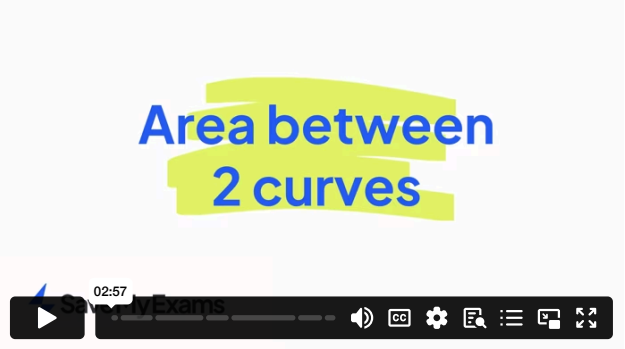
Updated on 16 February 2026



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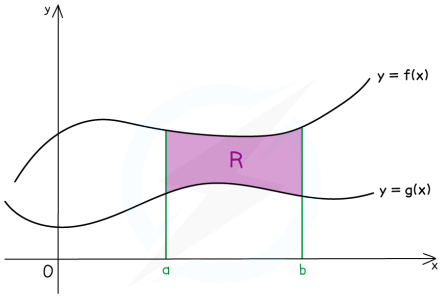
Area between 2 curves

What is the area between two curves?

- Ensure you are familiar with ...
  - Area under a curve
  - Area between a line and a curve



$$R = \int_a^b [f(x) - g(x)]dx$$

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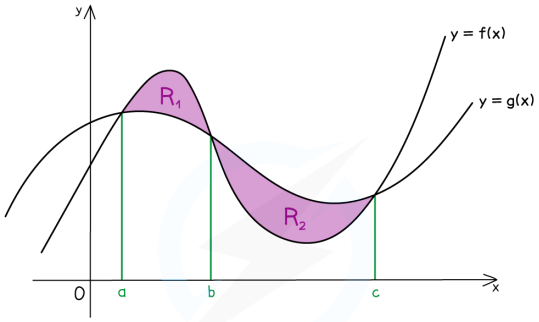
- In general find the definite integral of "upper curve" – "lower curve"
- However this does depend on ...
  - ... the area being found
  - ... if the curves intersect (and cross over)

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Help me understand



$$R_1 = \int_a^b [f(x) - g(x)]dx$$

$$R_2 = \int_b^c [g(x) - f(x)]dx$$

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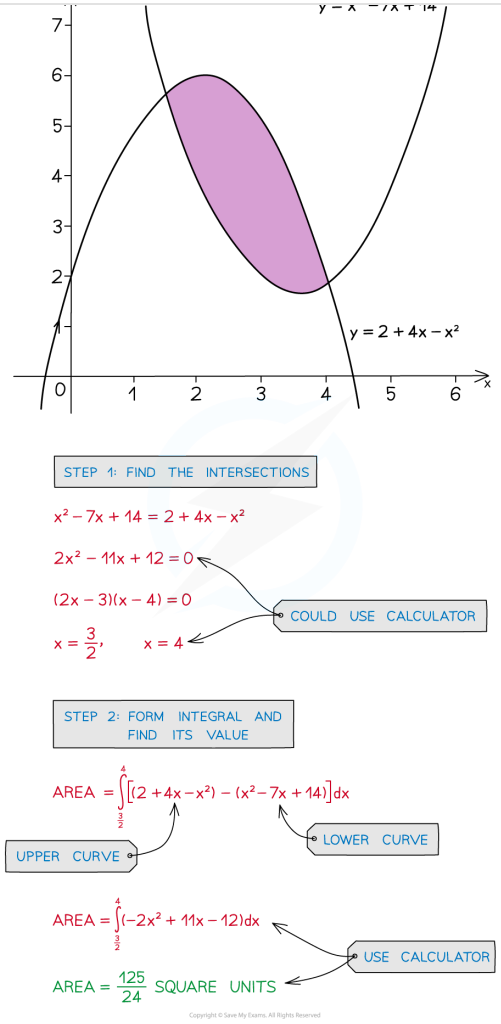
- The points at which curves intersect may need to be calculated

How do I find the area between two curves?



- **STEP 1**
  Find the intersections of the curves if needed
- **STEP 2**
  Form the integral …
  - … using the intersections as limits
  - … "upper curve" – "lower curve" …
  - … and find the value of the integral
- **STEP 3**
  Repeat STEP 2 if more than one area needed
- **STEP 4**
  Add areas together

Examiner Tips and Tricks

- If no diagram is provided sketch one, even if the curves are not accurate.
- Add information to any given diagram as you work through a question.
- Maximise use of your calculator to save time and maintain accuracy:
  - Solving equations, especially **cubic** equations

Worked Example

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